INCH-POUND

MIL-R-39016/606A
29 July 1994
SUPERSEDING
MIL-R-39016/606
24 August 1989

MILITARY SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 2 AMPERES (0.200-INCH TERMINAL SPACING), INCLUDING INSPECTIONS OF APPENDIXES A AND B

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.

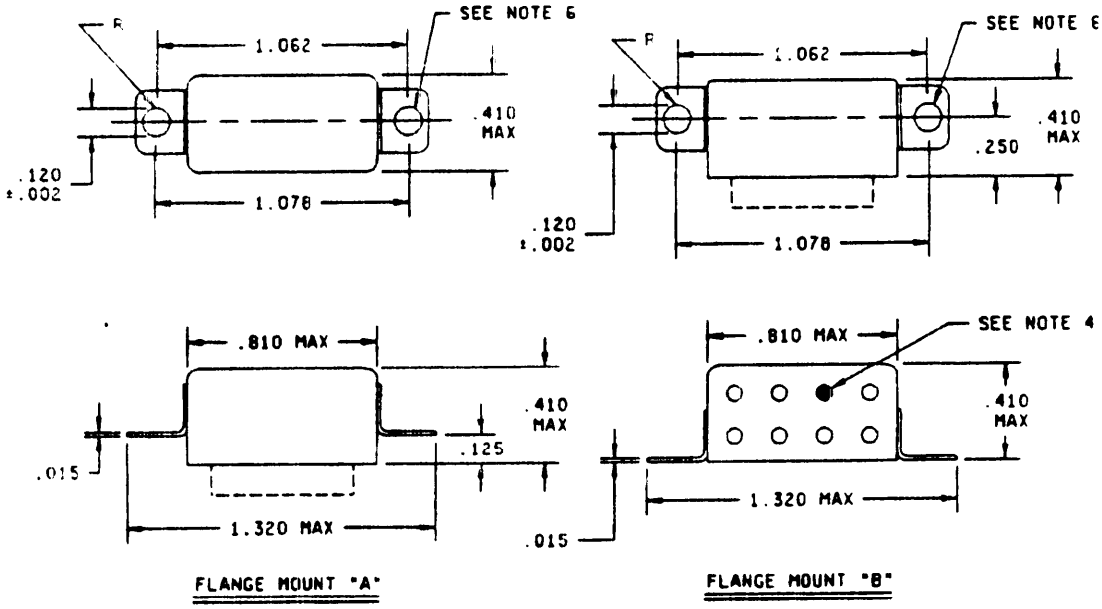
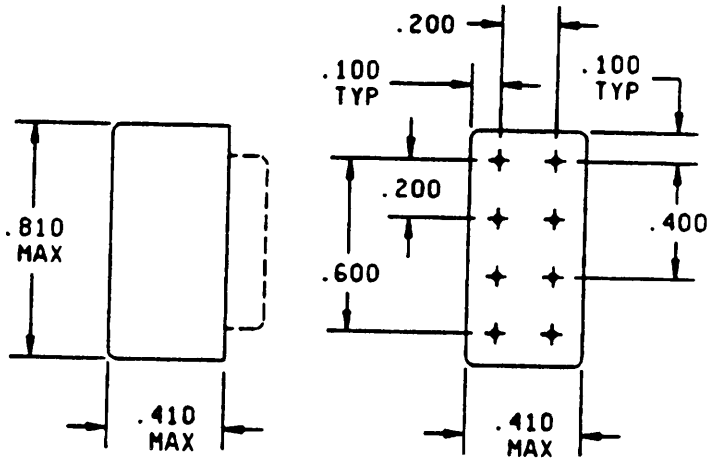


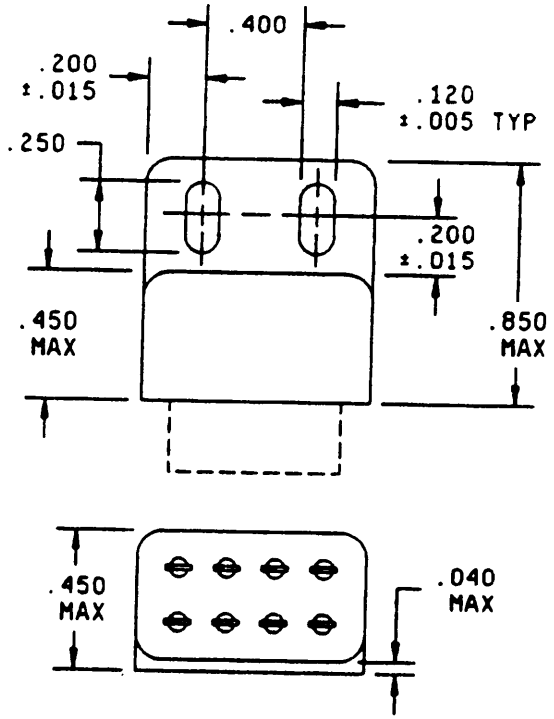
FIGURE 1. Dimensions and configurations.

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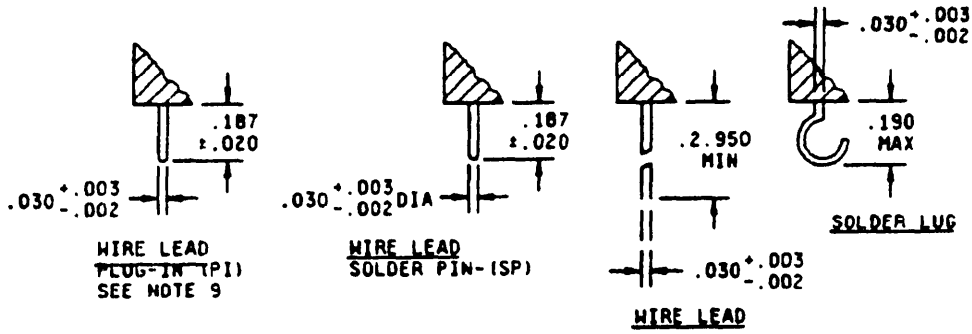
NO MOUNT

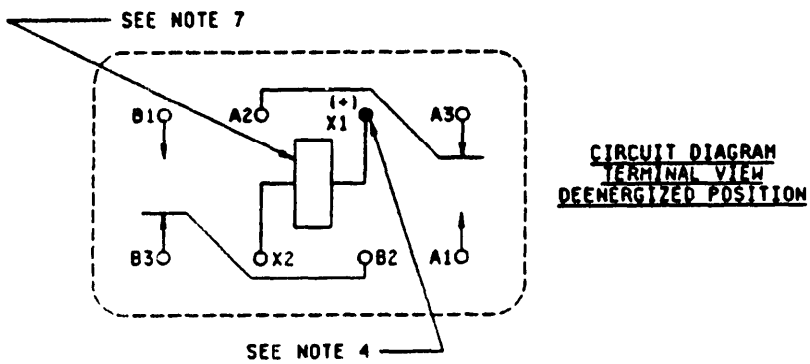


FLANGE MOUNT "C"

FIGURE 1. Dimensions and configurations - Continued.

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CIRCUIT DIAGRAM
TERMINAL VIEW
DEENERGIZED POSITION

Inches	mm	Inches	mm
.002	0.05	.250	6.35
.003	0.08	.400	10.16
.015	0.38	.410	10.41
.020	0.51	.450	11.43
.030	0.76	.600	15.24
.040	1.02	.810	20.57
.100	2.54	.850	21.59
.120	3.05	1.062	26.79
.125	3.18	1.078	27.38
.187	4.75	1.320	33.53
.190	4.83	2.950	74.93
.200	5.08		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Indicated terminal shall be marked with contrasting bead.
5. Circuit diagram (including terminal designations) marked on top if legible from the mounted position, otherwise marking surface is optional.
6. Mounting screw head clearance based on use of .112 (2.84 mm) diameter fillister head screws.
7. Coil symbol optional in accordance with MIL-STD-1285.
8. Mounting surface finish shall be compatible with aluminum (duralumin type) as defined by the compatible couples table of MIL-R-39016.
9. Finish gold plate in accordance with MIL-G-45204, type II, class I, grade C, except knoop hardness shall be 130 to 240, underplating nickel 50 microinches to 150 microinches thick.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Cleaning and small particle inspection: Applicable.

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:

2 amperes at 28 V dc.

.3 ampere at 115 V ac (60 Hz and 400 Hz case not grounded).

Inductive load: 0.75 ampere at 28 V dc with 0.200 henry inductance.

Lamp load: 0.160 ampere at 28 V dc.

Low level: 10 µA to 50 µA at 10 mV to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

High level:

Before life: 0.050 ohm maximum.

During life: Not more than 5 percent of open circuit voltage.

After life: 0.100 ohm maximum.

Low level:

Before life: 0.050 ohm maximum.

During life: 33 ohms maximum.

After life: 0.150 ohm maximum.

Intermediate current:

Before life: 0.050 ohm maximum.

During life: 1 ohm maximum.

After life: 0.100 ohm maximum.

Contact bounce: 2.0 ms maximum (applicable after high level rated load life and intermediate current tests).

Contact stabilization time: 2.0 ms maximum.

Overload (high level only): Two times rated resistive current. Not applicable to ac load ratings.

COIL DATA: See table I.

Operate time: 4.0 ms maximum over temperature range with rated coil voltage.

Release time: 4.0 ms maximum over temperature range from rated coil voltage.

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ELECTRICAL DATA:

Insulation resistance: 10,000 megohms minimum.

Dielectric withstanding voltage:

Sea level:

Between case, frame or enclosure, and all contacts; both in the energized and deenergized positions	1,000 V rms (60 Hz).
Between case, frame or enclosure, and coils	500 V rms (60 Hz).
Between all contacts and coils	1,000 V rms (60 Hz).
Between open contacts in the energized and deenergized positions	500 V rms (60 Hz).
Between open contacts	1,000 V rms (60 Hz).

Altitude:

Between all terminals to case	350 V rms (60 Hz).

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): Method 204, MIL-STD-202. Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts.

Vibration (random): Method 214, MIL-STD-202, test condition 1G. Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts. Applicable to qualification, group A, and group C inspection testing.

Shock (specified pulse): Method 213, MIL-STD-202, test condition B (100 g's). Contact chatter shall not exceed 10 µs maximum for closed contacts and 1 µs maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Particle Impact Noise Detection: Applicable.

PHYSICAL DATA:

Terminals: See figure 1 and table I.

Terminal strength: 3 pounds ±0.3 pound pull.

Terminal twist test: Applicable to wire lead terminals.

Solderability: Applicable.

Dimensions and configuration: See figure 1.

Weight: 13 grams (0.46 ounce) maximum.

Seal: Hermetic, adjunct sealant shall not be used.

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Minimum marking: Military Part or Identifying Number (PIN) "J" with the date code (example: J9237), circuit diagram, manufacturer's name or source code.

Outgassing: External materials (including marking ink) used in the finished relay shall be tested in accordance with ASTM-E-595. Samples to be tested shall have been processed in the same manner as that used in production of the qualification lot. Materials, when tested, shall not release greater than 1.0 percent total mass loss (TML) and 0.1 percent collected volatile condensable material (CVCM). Materials data listed in the latest revision of the NASA Reference Publication 1124 which meet these TML and CVCM requirements may be substituted in lieu of testing.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per contact load rating per relay.

Low level: 100,000 cycles per relay.

PART OR IDENTIFYING NUMBER (PIN): M39016/606- (plus dash number from table I and suffix letter designation failure rate level).

QUALIFICATION PROCEDURE: 1/

To obtain qualification to MIL-R-39016/606, a manufacturer shall:

a. Currently possess listing on QPL-39016 for all corresponding MIL-R-39016/6 dash numbers intended to be tested to MIL-R-39016/606 requirements.

b. Possess qualifying activity approved Particle Impact Noise Detection (PIND) and Cleaning and small particle inspection testing capabilities. At the discretion of the qualifying activity, approval may be granted based upon the following:

(1) An audit of the manufacturer's testing capabilities by the qualifying activity.

(2) The submission, to the qualifying activity, of procedures used in the performance of the PIND and Cleaning and small particle inspection tests.

(3) The submission, to the qualifying activity, of test data in accordance with the PIND and Cleaning and small inspection testing requirements specified in MIL-R-39016.

c. Submit to the qualifying activity, certification of intent to supply product qualified to MIL-R-39016/606.

1/ MIL-STD-790 requirements are applicable. Qualification at the initial failure rate level shall be granted based on corresponding MIL-R-39016/6 qualification. Retention of qualification and extension of qualification to lower failure rate levels shall be based on test data accumulated on MIL-R-39016/6.

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TABLE I. Dash number and applicable characteristics. 1/

Dash number 2/				Mount	Coil voltage (V dc) 3/		At +25°C				Over temperature range 4/		
Wire lead (PI) 5/	Solder lug	Wire lead (SP)	Wire lead		Rated	Max	Coil resistance ohms 10%	Specified pickup voltage (V dc) (max) 4/	Specified hold voltage (V dc) 4/	Specified dropout voltage (V dc) 4/	Specified pickup voltage (V dc) (max)	Specified hold voltage (V dc)	Specified dropout voltage (V dc)
001 008	002 005 009 012	003 006 010	004 007 011	A B None C	26.5	32.0	700	13.5	8.0	1.5	18.0	14.0	1.0
013 020	014 017 021 024	015 018 022	016 019 023	A B None C	12.0	15.0	160	6.4	4.0	0.7	9.0	5.8	0.50
025 032	026 029 033 036	027 030 034	028 031 035	A B None C	6.0	7.5	40	3.2	2.0	0.35	4.5	2.9	0.25
037 044	038 041 045 048	039 042 046	040 043 047	A B None C	5.0	6.0	27	2.7	1.65	0.29	3.8	2.4	0.21

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum, or peak ac open circuit, are not recommended for subsequent use in low level applications.

2/ The suffix letter M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 operations): M, 1.0; P, 0.1; R, 0.01. Example: 004M - - - - 004R.

3/ Caution: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

4/ Pickup, hold, and dropout voltages as shown are for test purposes only and are not to be used for design criteria.

5/ Solderability for plug-in relays is not applicable.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Navy - AS, MC, OS, SH
Air Force - 19, 99
NSA - NS

Preparing activity:
DLA - ES

(Project 5945-0918)

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